SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	2
Explanation	25%	2	2	2	2
Application	20%	1	1	1	1
Organization	15%	1	1	1	1
Accuracy	10%	1	1	1	1
Clarity	5%	1	1	1	1

Total Marks 32 /40

Signature of the Faculty

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests

(6)

CSA0710

Computer Networks

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32/40

) The presentation layer is the sixth layer of the Open System Interconnection model. It is responsible for data representation, translation & encryption. It ensure that data sent by one system can be interpreted by another system

Compression reduce the size of data before transmission while encryption protects data from unauthorized access Both operations are useful when transmitting large files over a network.

iven data

orginal file size = 900MB
Compression = 40%.
Encryption overhead = 5%.

Calculation

formula

$$\text{Compressed size} = \text{orginial size} \times \left(1 - \frac{\text{compression \%}}{100}\right)$$

$$= 900 \times \left(1 - \frac{40}{100}\right)$$

$$= 900 \times 0.6$$

Compressed size = 540MB

Encryption overhead $= \frac{5}{100} \times 540$

$= 0.05 \times 540$

Encryption overhead $= 27MB$

Encrypted size = Compressed size + overhead

$= 540 + 27$

Encrypted size $= 567MB$

Amount reduction $= \frac{900 - 567}{900} \times 100$

$= \frac{333}{900} \times 100$

Percentage Reduction $= 37\%$

Role of the layer

* Converts data into a suitable formate for communication between system
* Reduces file size and saves bandwidth
* Encryption: Protects sensitive infarmation from unauthorized access
* Helps ensure that information is interpreted correctly by the receiving application

Conclusion

The orginal file size of 900MB is reduced to 540MB after compression. The encryption overhead is 27MB, Giving a final encryted size of 567MB The total reduction compared with the original file is 37%. Therefore, compression improves efficiency. While encryption improves data security.

D Aim

To Calculate the total number of requests and the transmission time for a 3GB file using FTP and to explain how the application layer provides network services.

The Application layer is the seventh and topmost layer of the OSI Model. It provides services that allow user applications to communicate over a network

File transfer protocol is a Application layer protocol used to transfer files between a client and a server. It supports file uploading downloading and remote file access

The application is the problem divides a large file into smaller requests. The total number of requests & transmission time can be calculated using the given file size and network throughput.

Given

Total file size = 3GB

Effective throughput = 120Mbps

Request size = 3MB

Calculation

3GiB = 3 × 1024 MB

= 3072 MB

Number of Requests = $\frac{\text{Total file size}}{\text{Request size}}$

= $\frac{3072}{3}$

No. of Request = 1024

1024 Requests are Generated, each containing 3MB data

1 byte = 8bits

Total Bits = 3 × 1024 × 1024 × 1024 × 8

= 25,769,803,776

120Mbps = 120000000 bits / Second

Transmission Time = $\frac{\text{Total Data in Bits}}{\text{Throughput}}$

= $\frac{25,769,803,776}{120,000,000}$

Time = 214.75 Sec

Working of FTP

* The client establishes a connection with the FTP server
* The Client requests a file or initates a file upload.
* The server processes the request
* The receving computer stores the transfer file

Conclusion

The total file size is 3GB, which is divided into 1024 requests of 3MB each. At an effective throughput of 120Mbps, the estimated transmission time 214.75 sec. FTP enables file transfer between computers. while TCP supports reliable delivery

To calculate the total processing delay and end to end delay experienced by a packet and to explain the contribution of the OSI layers to network Communication

Introduction

The OSI Model divides network communication into seven layers. Each layer performs a particular function to ensure that data travels from the source to the destination

During transmission a packet experience different delays, processing delay occurs when network devices or protocols layers process data. Transmisson delay is the time needed to place packet bits onto a link Propagation delay is the time taken by the signal to travel through the communication medium

Given data

Application layer : 4ms

Transport layer : 3ms

Network layer : 5ms

Data link layer : 2ms

Physical layer : 1ms

propagation delay : 18ms

Transmission delay: 12ms

Calculation

Formula

$$D_p = \sum D_{processing}$$

$$D_p = 4+3+5+2+1$$

$$D_p = 15ms$$

$$D_{total} = D_p + D_t + D_{prop}$$

$$= 15+12+18$$

$$D_{total} = 45ms$$

Transmission delay

It is the time required to place all packet bits onto a communication link. It depends on packets size & transmission rate

$$D_t = \frac{\text{Packet size Bits}}{\text{Transmission Rate}}$$

Conclusion

The total processing delay is 15ms the transmission delay is 12ms and the propagation delay is 18ms

∴ The total end to end delay experienced by the packet is 45ms under the assumptions given in the Question.

9.) To Calculate the total number of frames total protocol overhead, total data transmitted and transmission effeciency for a network transfer 80MB of useful data

Given data

Useful data = 80MB
Useful data per frame = 1600 Bytes
Protocol overhead = 80 bytes.

of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation

Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand